

2023 NATIONAL YOUTH RISK BEHAVIOR SURVEY

Co-Occurring Adverse Experiences & Protective Factors in Adolescent Suicidality

A survey-weighted analysis of self-reported
adversity, protective behaviours, and suicide-risk
outcomes among U.S. high school students



TEAM 04 — “THE THERAPISTS”

Jacinta · Aliya · Destiny · Albert · Terence · Francis · Bernard



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01

Introduction

An introduction to our project,
key aims and objectives

Background



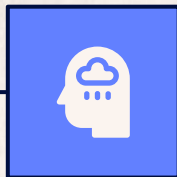
2021



Emergency

American Academy of Child and Adolescent Psychiatry declared a "national emergency in child and adolescent mental health" in the U.S.

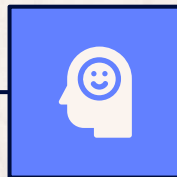
2022



Hearing

The U.S. Senate HELP Committee held a formal hearing, *"Why Are So Many American Youth in a Mental Health Crisis?"*

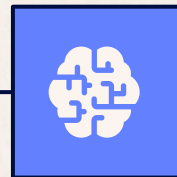
2023



Evidence

CDC's national survey quantifies the scale of the crisis among U.S. high schoolers

2026



Investigation

Our project examines which adverse and protective experiences co-occur with this risk



Background

Suicide is the second leading cause of death among Americans aged 10–34

Over the past decade, rates of serious suicidal ideation have rise

Background

1 in 5

seriously considered attempting
suicide in the past 12 months

39.7%

of students reported persistent
sadness or hopelessness
in the past 12 months

9.5%

attempted suicide one or more times
in the past 12 months

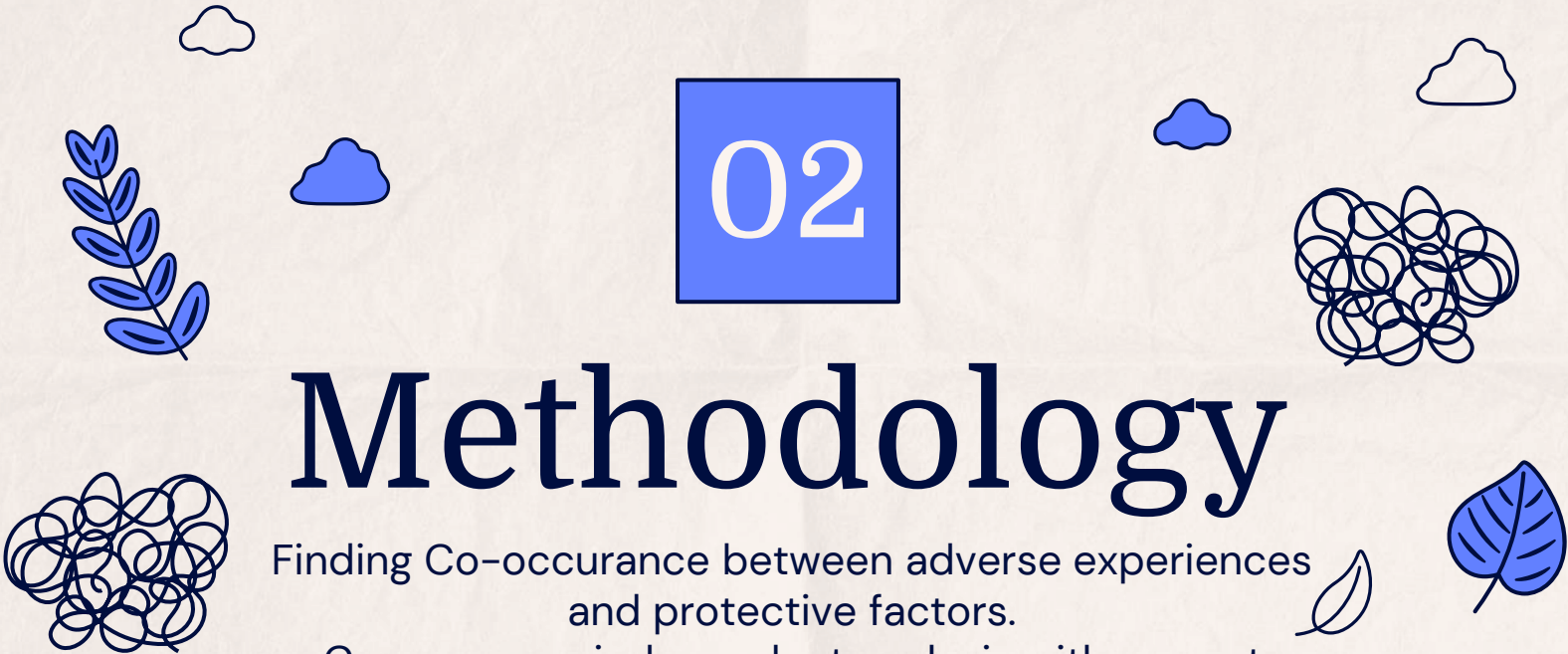




Learning More

Suicide risk rarely comes from one thing.
This gap motivates our research question:

How do adverse and protective experiences co-occur with adolescent suicidality, and what does that mean for where prevention efforts should focus?



02

Methodology

Finding Co-occurrence between adverse experiences
and protective factors.

Compare our independent analysis with current
research

Study design: 2023 National YRBS



What is it?

A nationally representative, biennial CDC survey of U.S. high school students (grades 9–12)

Anonymous, self-report questionnaire: every value in this dataset is what a student said, not something independently verified

20,103 usable student responses in the 2023 cycle measuring over 100 different variables.

Three-stage stratified cluster design:
strata → primary sampling units (PSUs, roughly clusters of schools) → classrooms



Why does this design matter?

Weight

Each respondent stands in for a number of real students nationally (not every student in the survey represents just one person)

stratum / psu

Students in the same school answer more alike than random students elsewhere.

Ignoring this understates uncertainty.

Our Selected Variables

	Domain	Details	Variables
Jacinta	Demographics	Age, Ethnicity etc	Q1, Q2, Q3, raceeth
Aliya	Protective factors	Grades, Sleep, Basic Needs Met, Social life	QN85, QN87, QN99, QN103, QN104
Destiny	Adverse factors	Sexual dating and violence	QN19, QN20, QN21, QN88
Albert	Adverse factors	School and community	QN22, QN23, QN24, QN25, QN86
Terence	Adverse factors	Household dysfunction	QN89, QN90, QN91, QN100, QN101, QN102
Francis	Risk behaviours	Violence and mental distress	QN12, QN13, QN16, QN26, QN84
Bernard	Risk behaviours	Substance use	QN43, QN48, qntb4, qnillict, qnowt, qnobese

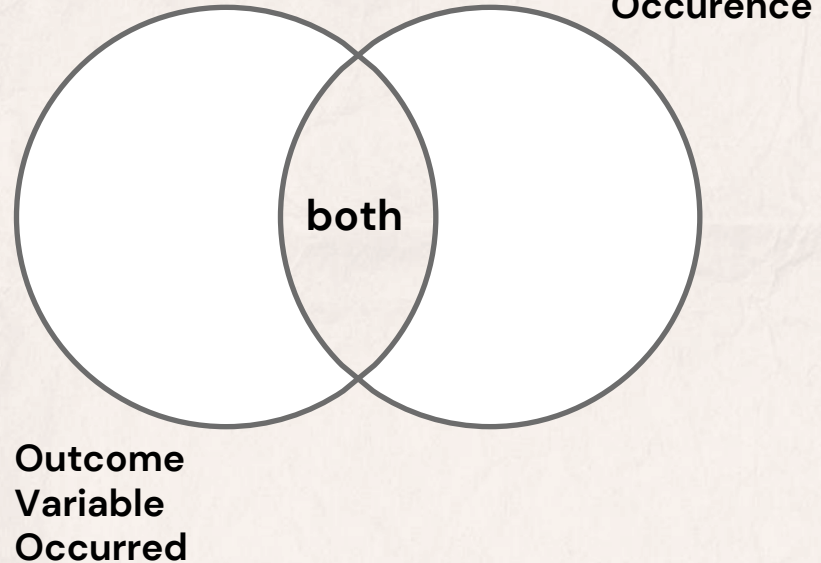
Variable Research, with cited Publications for the inclusion on each factor

Outcome Variables

"QN27", # seriously considered suicide (12mo)
"QN28", # made a suicide plan (12mo)
"QN29", # attempted suicide 1+ times (12mo)
"QN30" # attempt resulted in injury

p-values, why not?

118 out of 124 were significant; as such, the p-values showed limited ability to distinguish important from negligible associations.



Reproducibility via GitHub

Everything lives in one place

All scripts, data, documentation, and outputs are tracked in a single public repository to see the full history of the project

Every change is logged and attributed

21 commits across the project, each with a description of what changed and who changed it (from variable selection to charts)

Structured for reproducibility

Separate folders for data, scripts, output, markdown, and documents. The pipeline from raw data to final figures is traceable end-to-end.

Analysis can be re-run from scratch

Anyone can rebuild our results because the R scripts and outputs are version-controlled together

Commits

History for BIP-Technical-Challenge-Collaboration / documents on [main](#)

Commits on Aug 17, 2026

Risk Behaviours

 FrancisAsane authored 3 days ago

Add files via upload

 ali-pixel545 authored 3 days ago

Commits on Aug 16, 2026

Add files via upload

 ajsastheticjourneys authored 4 days ago

Commits on Aug 14, 2026

Updated to include QNILLICIT variable

 bernkyl authored last week

Add files via upload

 Terence03-art authored last week

Commits on Aug 12, 2026

Add files via upload

 ajsastheticjourneys authored last week

Commits on Aug 9, 2026

Add files via upload

 cakemaster1017 authored 2 weeks ago

Add files via upload

 ali-pixel545 authored 2 weeks ago

Commits on Aug 5, 2026

Variable Research

 bernkyl authored 2 weeks ago

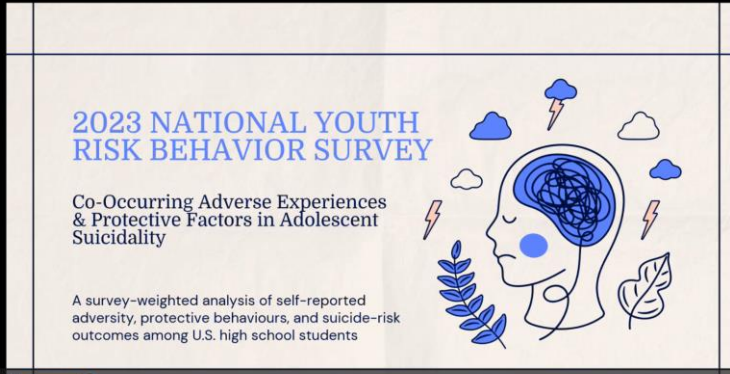
Collaboration via Google Drive & Teams

Shared with me > BIP Technical Challeng... > Recordings ▾

◆ Ask Gemini | Type ▾ People ▾ Modified ▾ Source ▾

Name ↓ Owner More info (Option + →) Date modified

 Wednesday Meeting Team 04.-20260729_141217...		 chukwuedojacinta	29 Jul
 Team 04 Meeting.-20260802_192840-Meeting R...		 chukwuedojacinta	
 Team 04 Meeting.-20260726_190616-Meeting Re...		 chukwuedojacinta	



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DI
Destiny Umanah (Stu...)

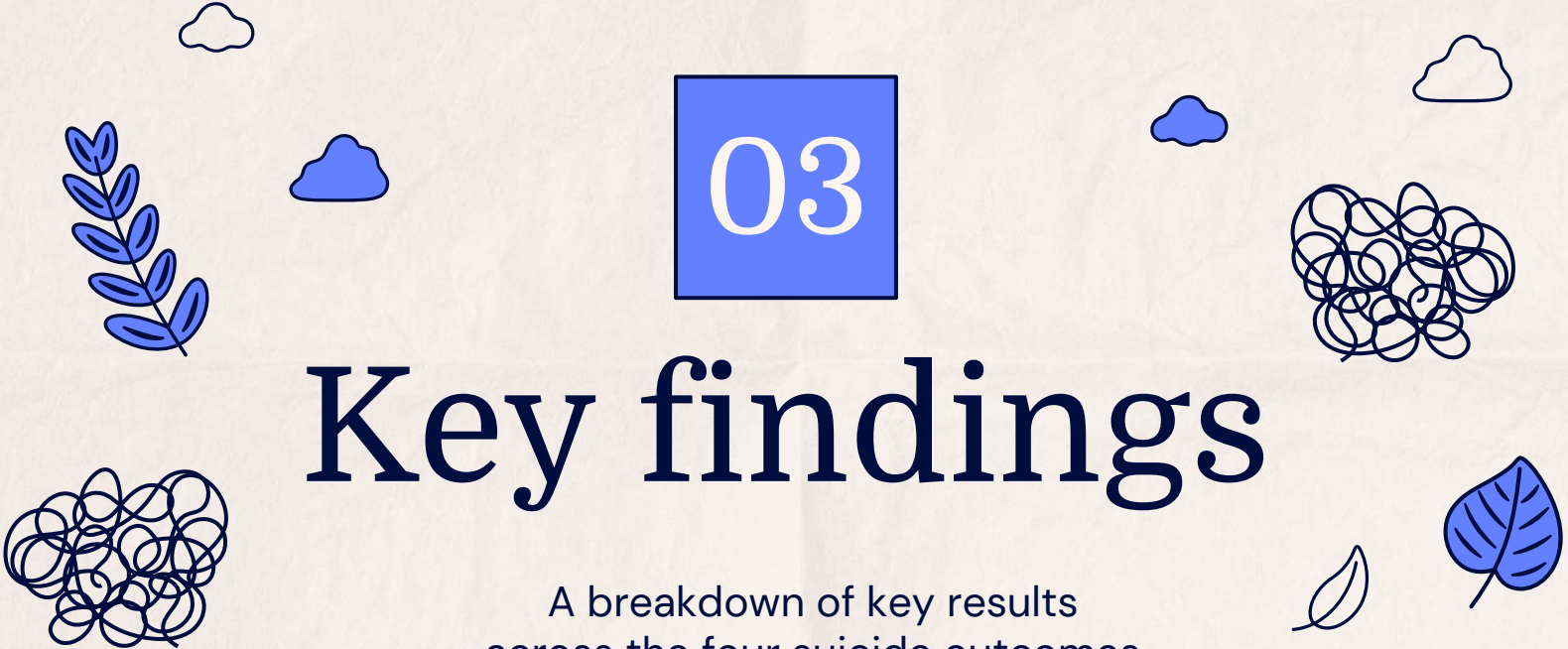
BK
Bernard Kymoni

AO
Aliya Umanah (Stu...)

FA
Francis Umanah (Stu...)

00:02:39 Jacinta · Aliya · Destiny · Albert · Terence · Francis · Bernard

HDR UK BIP Presentation Demo! - 20260820_19...



03

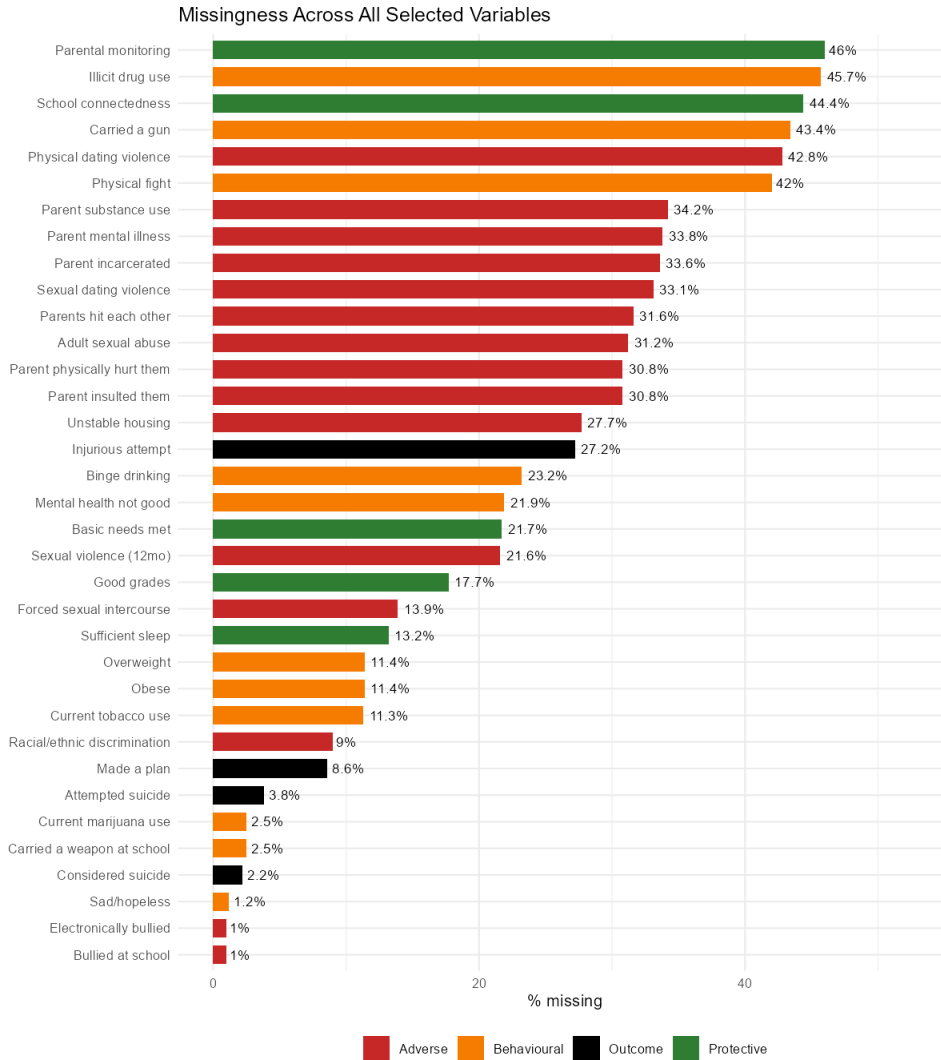
Key findings

A breakdown of key results
across the four suicide outcomes

Variables Missingness Audit

Missingness ranges from 1.0% to 46.0%, with the highest-missingness variables found in the household-adversity, connectedness, and monitoring cluster

CDC's minimum-cell-size convention (unweighted N $\geq$ 30 per reported group) was applied throughout the analysis



Weighted prevalence: Point estimate with 95% CI

Adverse Behavior Outcome Protective



Weighted Prevalence of Variables

Weighted prevalence across the variables ranges from 2.0% (parents hit each other) to 86.3% (basic needs met)

Four of the five Protective factors exceed 50%: Protective factors are the norm in this population, not the exception

These estimates describe how common a given experience or behaviour is in the **population**



Demographics (Jacinta)



Suicidality Outcomes by Sex

Weighted % with 95% CI, all four outcomes

Female Male

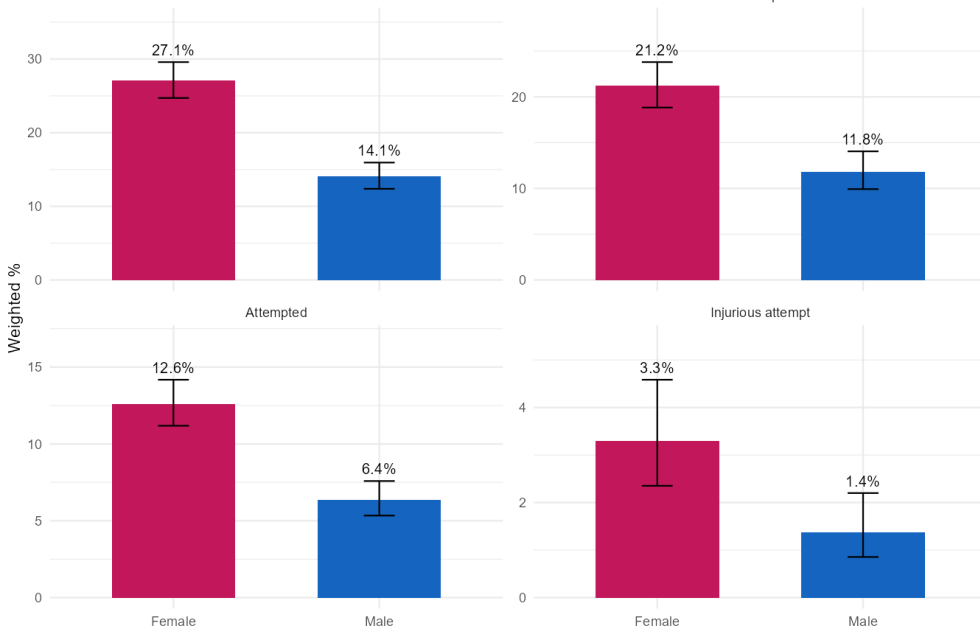
Considered suicide

Made a plan

Attempted

Injurious attempt

Weighted %



Outcomes Prevalence by Sex



Sex shows the strongest and most consistent demographic pattern in this dataset

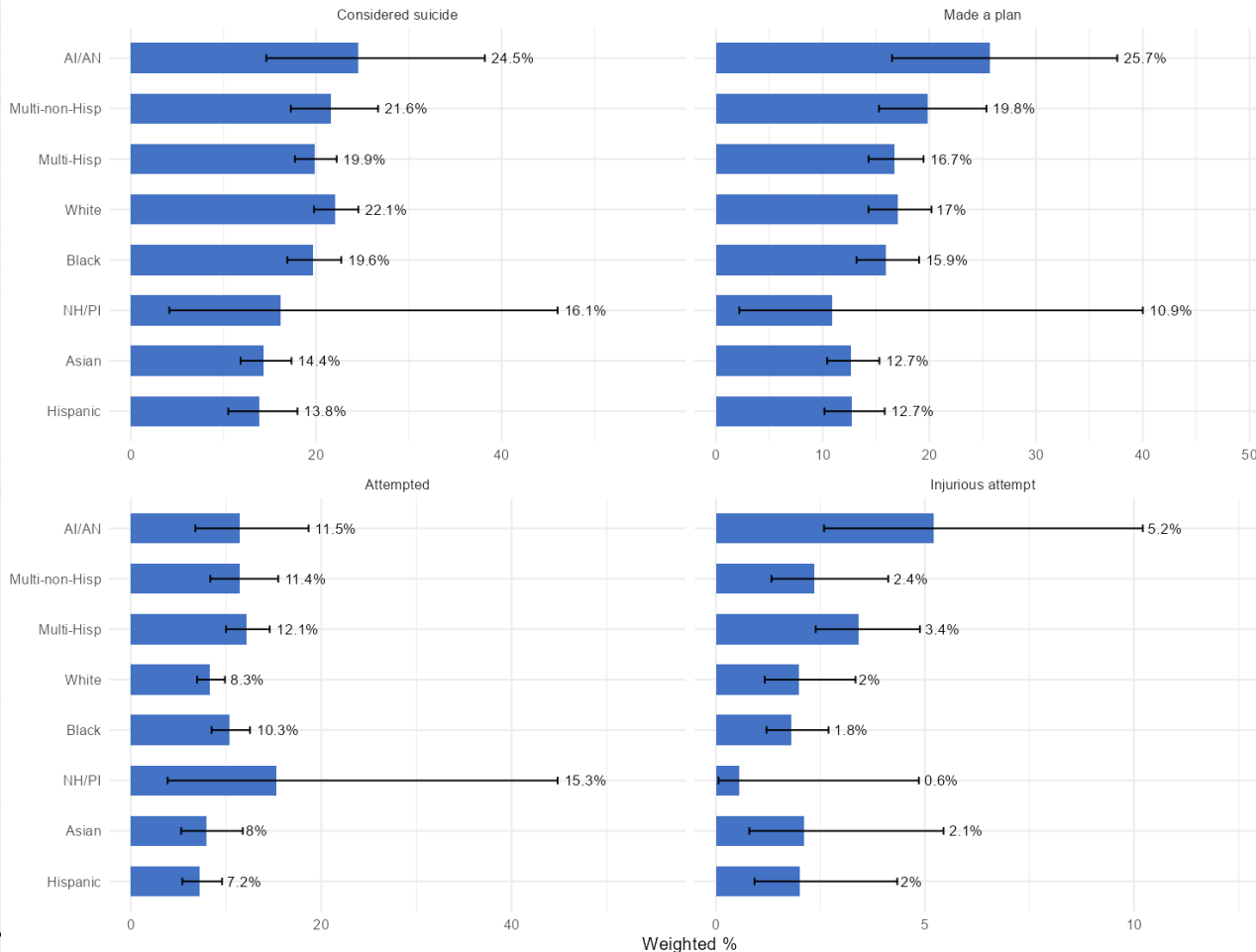
Female students report 2x the prevalence of male students across all four outcomes

Respondents Racial Composition

- White 48.1%
- Multi-Hispanic/Latino 19.8%
- Black 13.3%
- Hispanic 7.7%
- Multi-non-Hispanic 6.1%
- Asian 4.3%

Suicidality Outcomes by Race/Ethnicity

Weighted % with 95% CI, all four outcomes



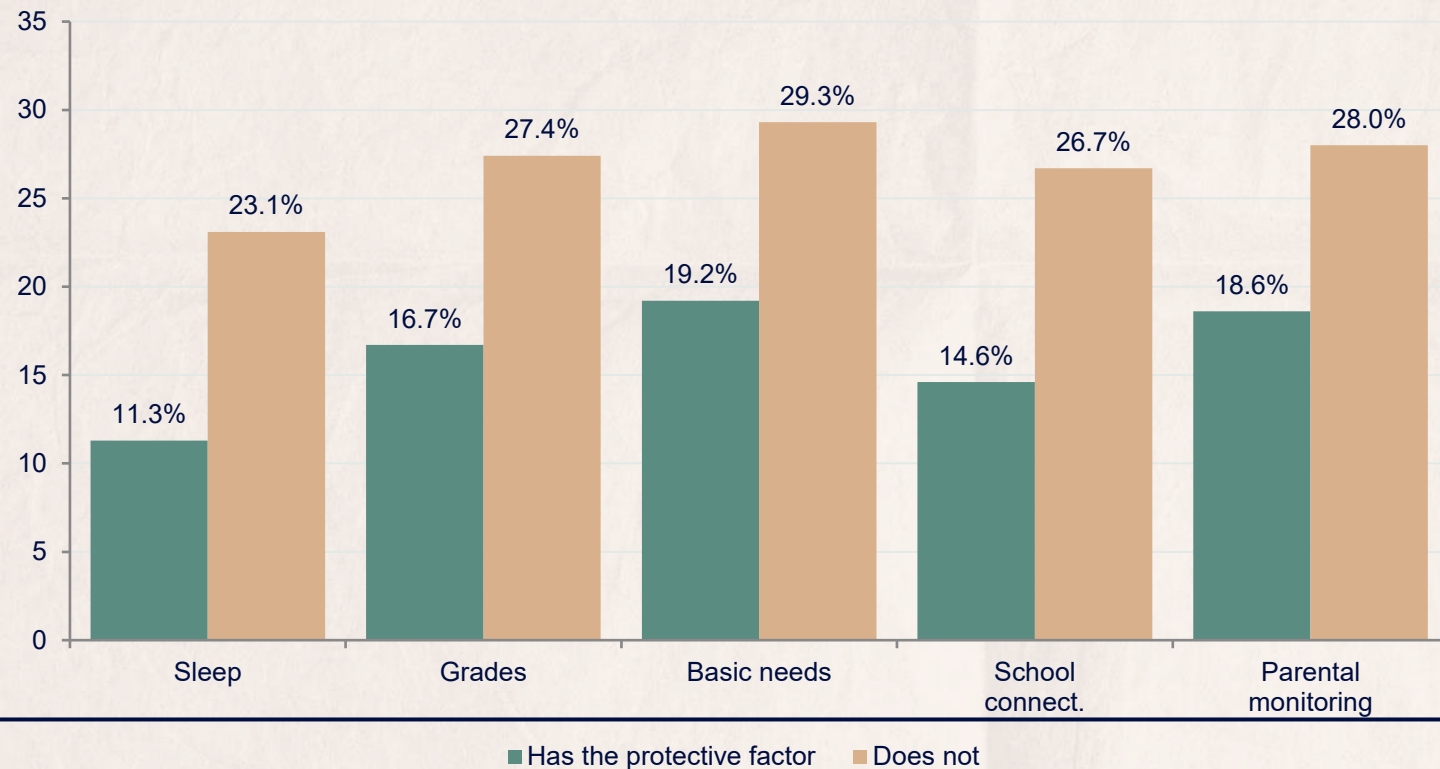
Outcomes Prevalence by Race/Ethnicity

Race/ethnicity is statistically significant for considered suicide and attempted suicide ($p < 0.05$), but not significant for made a plan or injurious attempt ($p > 0.05$)

AI/AN students generally had the highest or near-highest prevalence. Wider confidence intervals for AI/AN and NH/PI groups reflect smaller subgroup sample sizes

Protective Factors (Aliya)

% who seriously considered suicide (QN27), by protective factor



**HOLDS ACROSS
ALL OUTCOMES**

All 5 protective factors showed lower prevalence on **every one** of the 4 suicidality outcomes:

1. Considered
2. Planned
3. Attempted
4. Injured



Protective Factors Validated By CDC's Findings



Factor	CDC adjusted PR (2023 YRBS)	Our ratio (unadjusted)	Agreement
Sleep (QN85)	0.53 - 0.67	0.48 - 0.49	Close match
Basic needs met (QN99)	0.41 - 0.80	0.23 - 0.66	Close match
Parental monitoring (QN104)	0.51 - 0.74	0.34 - 0.66	Close match
School connectedness (QN103)	0.63 - 0.70	0.52 - 0.56	Close match
Grades (QN87)	Not tested by CDC	0.34 - 0.66	Consistent with outside literature

A ratio below 1.0 = lower risk in the protective-factor group. 0.53 $\approx$ roughly half the risk.

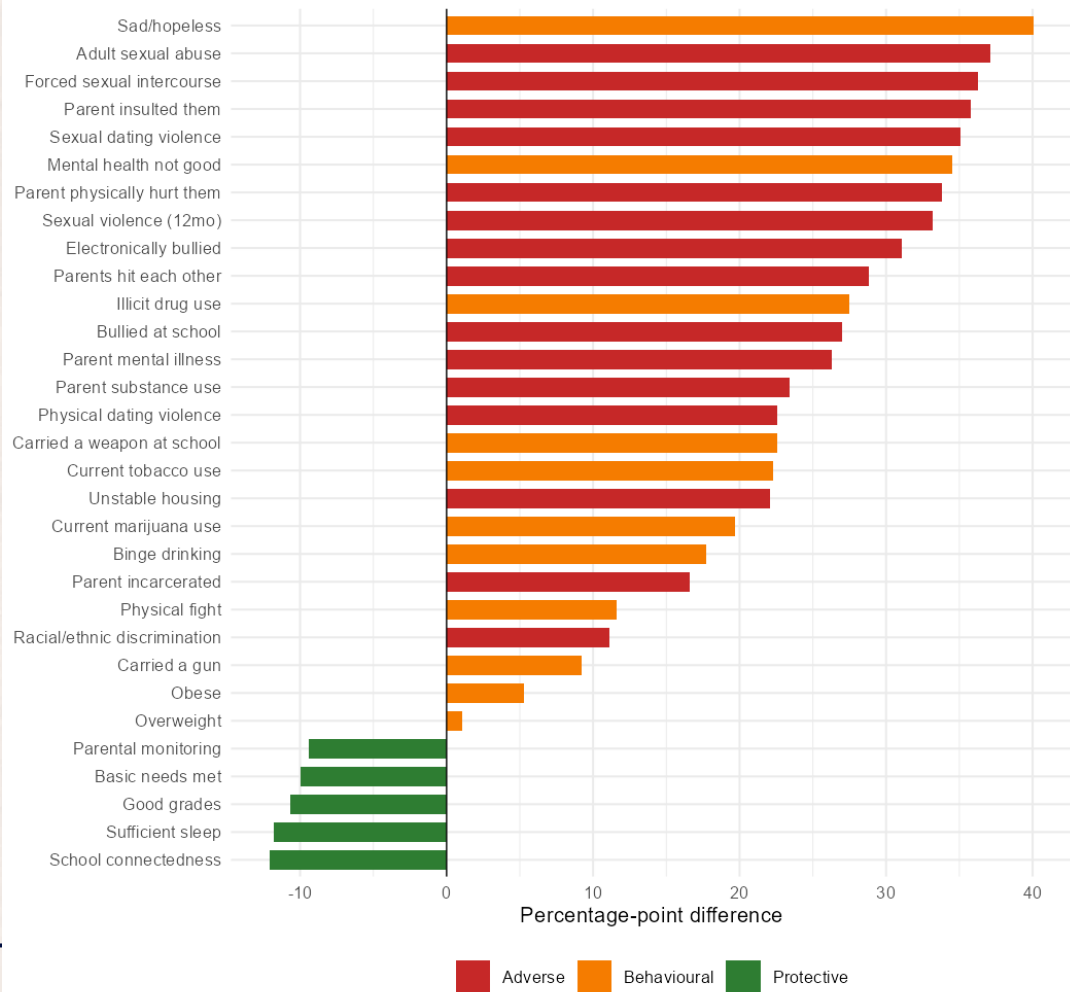
Sleep, basic needs, parental monitoring and school connectedness are the exact protective factors CDC selected for its own 2023 YRBS analysis: and our unadjusted ratios sit close to their adjusted ones.

Grades isn't one of CDC's six protective factors, so we drew on outside literature instead — directionally consistent, but the least consistently supported of the five across subgroups in the wider research.



Outcome-Selected Variables(Predictors) Comparison

Considered suicide

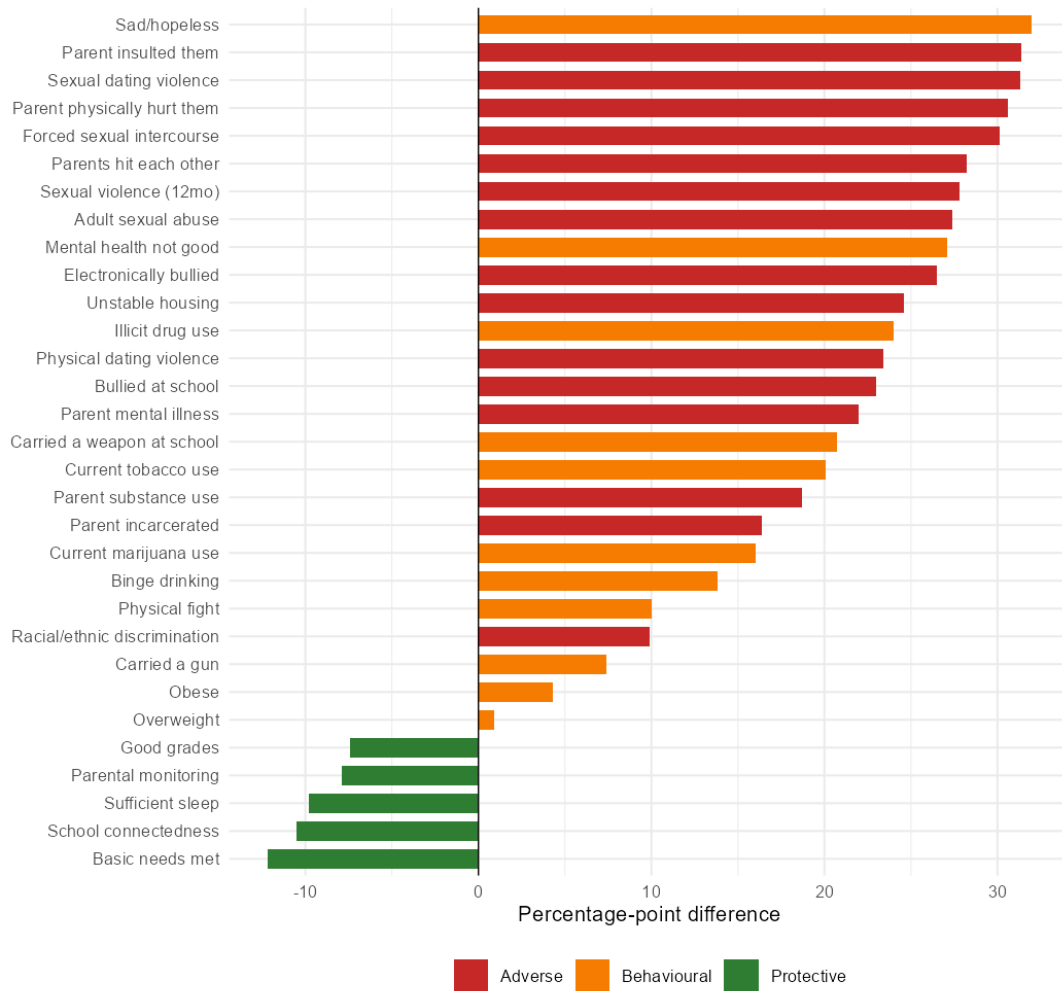


Cooccurrence

- Sad/Hopelessness
- Adult Sexual Abuse
- Forced Sexual Intercourse
- School connectedness
- Sleep
- Basics Needs Met



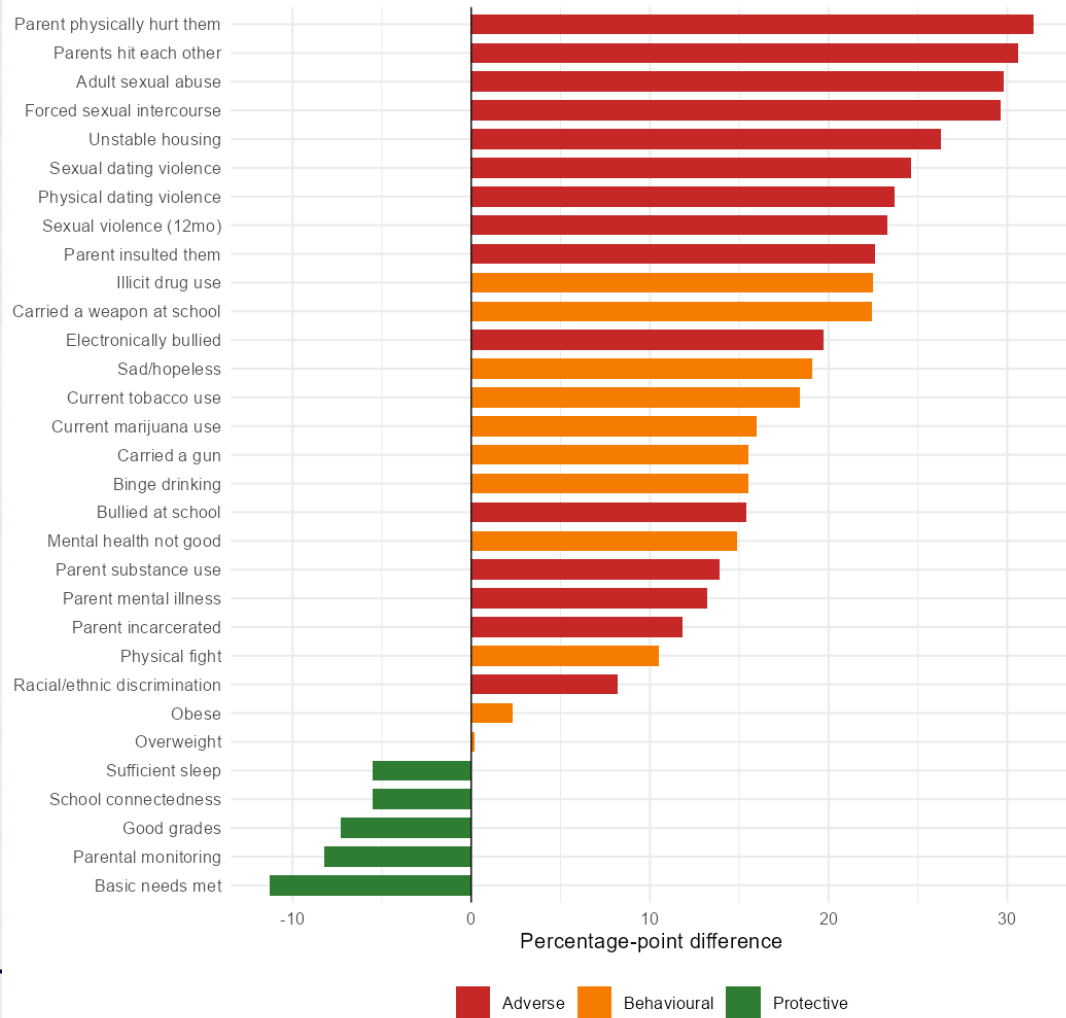
Made
a plan



Cooccurrence

- Sad/Hopelessness
 - Parents Insulted Them
 - Sexual Dating Violence
- ↑
- Basic Needs Met
 - School Connectedness
 - Sufficient Sleep
- ↓

Attempted

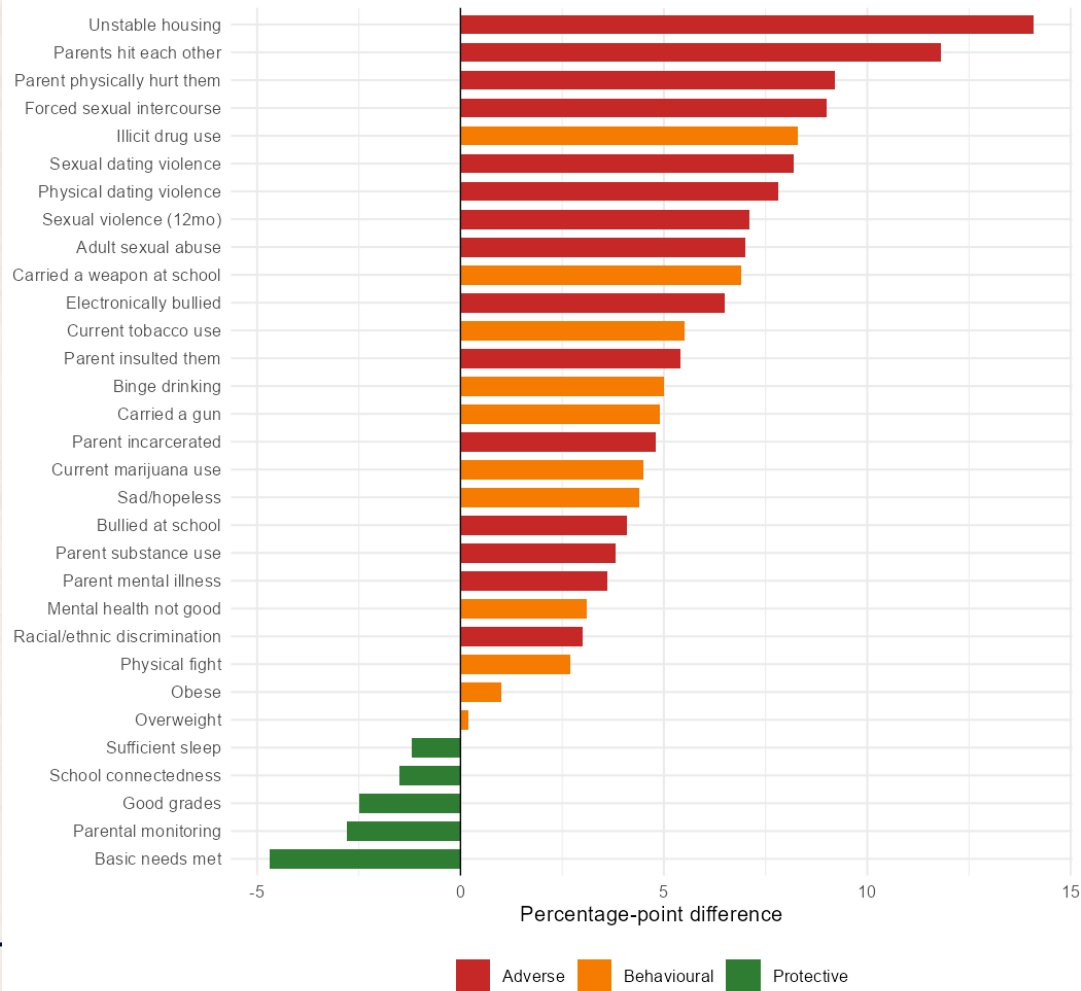


Cooccurrence

- Parent's physically hurt them
- Parents hitting each other
- Adult Sexual Abuse
- Basic Needs Met
- Parental monitoring
- Good Grades



Injurious attempt



Cooccurrence

- Unstable Housing
- Parent's hitting each other
- Forced Sexual Intercourse
- Basic Needs Met
- Parental Monitoring
- Good Grades





04

Limitations

A closer look at our dataset
refinements and further analysis

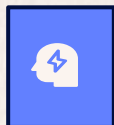


Limitations



Cross-Sectional, Self-Reported Data

Findings indicate co-occurrence, not causation or prevention.



Missing Data

Per-comparison exclusion was not tested against alternative approaches, so the sensitivity of findings to missing-data assumptions is unknown.



Variable Coverage

Sexual orientation and gender identity were outside the scope of the analysis, while the null association for weight status may reflect the absence of measures capturing perceived weight and weight-based teasing.



Unmeasured Confounders

Household socioeconomic status is known to shape both protective factors and mental health outcomes; however, it isn't captured in the YRBS.



Limitations cont'd



Demographic Subgroup Uncertainty

American Indian/Alaskan Native, Native Hawaiian/Pacific Islander, and the youngest age/lowest grade groups had wider confidence intervals, reflecting smaller sample sizes.



Recall & reference periods

Different questions ask about “during your life,” “past 12 months,” and “past 30 days”. These inconsistent windows can affect recall and comparability.



Overlapping factors

Protective factors are interrelated (i.e., a well-monitored student may also sleep more), so this analysis can't isolate each factor's independent contribution.

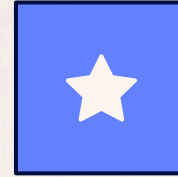


05

Conclusions

A summary of our future
direction and next research
question

Two sides of the same picture



Adverse experiences

- Risk shifts with the severity of outcomes, not one factor throughout
- Distress and sexual violence lead ideation and planning outcomes
- Household violence and unstable housing take over at attempt and injury

Protective factors

- Protective factors are common in the population, not rare
- Their absence, not their scarcity, tracks with worse outcomes
- Basic needs met is the most consistent factor across the whole pathway

The largest adverse effect always outweighs the largest protective one, and the gap narrows for the more severe outcomes.



Future Directions

Robustness testing

Re-estimate key findings under alternative missing-data assumptions, such as complete-case analysis



Multivariable modelling

Adjust for demographic and other potential confounders, consistent with the CDC's companion analysis

Expanded equity analysis

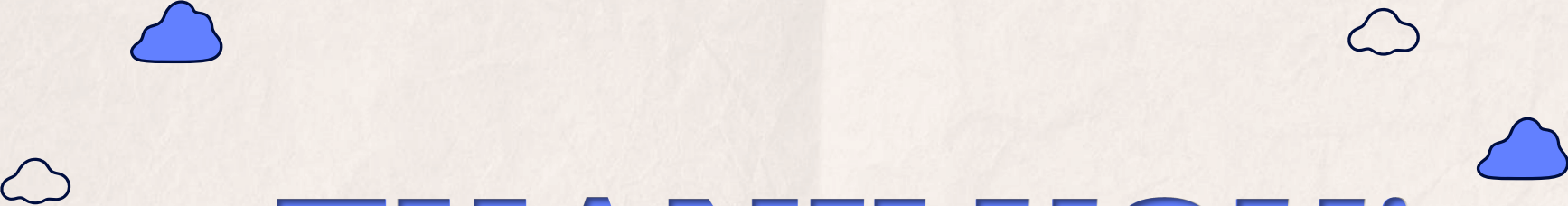
Incorporate sexual and gender identity using the raw survey data and appropriate recoding



Next Research Question?

Do the strength of key co-occurrence associations, rather than only outcome prevalence, vary across sex and racial/ethnic groups?

For example, does the association between sadness/hopelessness and suicidal ideation remain consistent across demographic subgroups, or is it concentrated within populations?



THANK YOU!

Team 04 – “The Therapists”

Questions?
